

Algorithms Overview (1 pt)

Consider the task of optimizing a **large-scale** problem with $n = 10^6$ variables, given that state-of-the-art implementations of the following methods are available:

- (a) Gradient Descent method
- (b) Newton's method as on Page 12, Slides-07
- (c) Augmented Lagrangian method using Newton's solver
- (d) Infeasible Start Newton's method
- (e) Active Set Method using Newton's solver
- (f) Primal-Dual Interior Point method using Newton's solver with Hessian modification
- (g) Gauss-Newton method
- (h) L-BFGS method

For every of the following scenarios, list algorithms that can find a local minimum and specify the algorithm for which you expect the shortest runtime (assuming infinite memory). Note that every problem starts from a feasible point. **Briefly justify your choice.**

1. A quadratic problem which is strictly convex with linear equality constraints.

In this case c, d, e and f would be able to find a local minimum. Not the others due to the equality constraints. d, e and f would be the fastest because c needs to tune dual variables and the penalty.

3. A convex quadratic problem with linear inequality constraints which can be solved via a sequence of convex quadratic subproblems with small number of linear equality constraints.

The Active Set Method (e) would be the suitable algorithm because it automatically generates the sequence of subproblems by identifying the active constraints.

5. A non-convex least-squares problem of the form $f(x) = \sum_{i=1}^m f_i(x)^2$ with $x \in \mathbb{R}^n$. The gradient of f_i is dense and the hessian is not available.

As the hessian is not available, only methods that don't need the Hessian or second order information will work. Then a, g and h would work. h would be the best because the gradient is dense and thus the approximated Hessian will be dense too.

2.) A linear program with inequality constraints $\rightarrow$ Hessian is 0

SOL: - Augmented Lagrangian method using Newton's solver $\rightarrow$ quadratic penalty term makes Hessian non-zero

- P-D Int. Point method $\rightarrow$ Projected Hessian becomes pos. definite.

• Shortest runtime: P-D. Int. Point method $\rightarrow$ less iterations, no outer loop

4.) An unconstrained non-convex problem

SOL: - Gradient Descent

- P-D Int. Point method

- L-BFGS method

- Gauss-Newton if obj. func $:= \sum f_i(x)^2$

• Shortest runtime: P-D Int Point method $\rightarrow$ Newton's solver is fastest.